

Computer Vision System for Briscola Round Analysis

Group 2R8\$

Final Project Proposal

1 Introduction and Motivation

Briscola is a traditional Italian card game played with a 40-card regional deck. During a match, keeping track of the points collected by each player is essential, but it can be difficult because point values are not ordered in the same way as card ranks and because the score is accumulated over many rounds. The goal of this project is to develop a computer vision system that analyses short videos of Briscola rounds and automatically extracts the information needed to assign points and update the game score.

The proposed system focuses on the two-player version of the game. For each round, the input is a short video clip showing the full action of the round, from the table state before the play to the moment when both players have played and collected the cards.

The expected output is a structured description of the round: the two played cards, the card defining the trump suit (*briscola*), the player associated with each card, the player who plays first, the winner of the round, and the points assigned to that winner. By applying the system round by round, the final score of an entire game can be reconstructed.

2 Task Definition

For each input video corresponding to one round, the system should:

1. detect and localize the visible cards on the table;
2. recognize the rank and suit of the two cards played in the round;
3. determine the briscola card used in the round, normally by recognizing the visible briscola card, but in the last videos of the game it will not be present;
4. associate each played card with the correct player: in the annotations, North and South denote the two fixed directions from which the cards come from (Above = North , Below = South) ;
5. determine the round winner;
6. compute the number of points collected in the round and update the cumulative score.

The main output for each round should follow the provided CSV format, where rank and suit are stored in separate fields:

```
Round, North_Number, North_Suit, South_Number, South_Suit,  
Briscola_Number, Briscola_Suit, Leader, Winner, Points.
```

The final output of a game should be provided in this form : the final totals of the accumulated points for each player and the overall winner.

3 Brief Game Rules

For this project, only the rules needed to assign points are relevant. In each round, one player leads by playing the first card and the other player answers with one card. The winner is determined as follows: If a card that has the same suit as the briscola card is played, the highest ranking briscola card played wins. If the card that answers the first one played is of different suit than the first one, the first card played wins. Else, if both cards have the same suit, the highest ranking card wins.

The card ranking, from strongest to weakest, is:

$$1 > 3 > 10 > 9 > 8 > 7 > 6 > 5 > 4 > 2$$

The point value assigned to each card is shown in Table 1.

The winner of the round receives as many points as the sum of the points of the two cards played in the round.

Card	1	3	10	9	8	7	6	5	4	2
Points	11	10	4	3	2	0	0	0	0	0

Table 1: Card ranking and point values.

4 Dataset

Dataset repository: [Google Drive folder](#)

The dataset contains recordings of Briscola rounds, reference card images, and ground-truth annotations. The folder `Briscola_Trentine` contains 40 scanned and labelled card images, one image per card, of the same Trentine deck used in the videos. All the cards used in the videos are in this image dataset.

The dataset contains four complete games, each with 20 short videos, for a total of 80:

Subset	Clips	Notes
<code>game1</code>	20	controlled acquisition, clear pauses before card collection
<code>game2</code>	20	controlled acquisition, similar setup with different lighting condition
<code>game3</code>	20	more realistic play, with hands, occlusions, overlap, and less stable timing
<code>game4</code>	20	controlled acquisition, different background

Ground truth for each game is provided in CSV files such as `game1results.csv`, `game2results.csv`, `game3results.csv`, and `game4results.csv`. The CSV files contain one row per round with the exact fields listed above: the round number, the number and suit of the card played by North, the number and suit of the card played by South, the number and suit of the briscola card, the leading player, the round winner, and the points assigned in that round. A corresponding text file, such as `game1output.txt`, stores the same round information in a readable format and additionally reports the final game winner and the total points for both players. The final result of the game is only reported in the `.txt` file.

5 Expected Challenges and Simplifications

The main visual challenges are small card details, rotated cards, similar suits, patterned tablecloth backgrounds, partial card overlap, hands entering the scene, and the presence of irrelevant card stacks. The hardest subset is expected to be the realistic game, where cards may be partially hidden and there may be no clean pause between playing and collecting the cards.

The following simplifications are provided:

- the deck type is fixed to the Trentine cards provided as reference images;
- the camera viewpoint is fixed above the table for the whole game.

6 Performance Measurement

Performance is evaluated over one complete game by comparing the predicted outputs with the manually annotated ground truth. Therefore, all 20 rounds of the game must be analyzed before computing the final metrics. The evaluation is divided into card recognition, player identification, briscola recognition, and final result prediction.

- **Card recognition accuracy:** measures the proportion of played cards correctly recognized across the game. A card prediction is considered correct only if both rank and suit match the ground truth. Since two cards are played in each of the 20 rounds, the metric is defined as:

$$A_{\text{card}} = \frac{N_{\text{correct cards}}}{40}$$

- **Player identification accuracy:** measures the proportion of correctly identified player roles across all rounds. For each round, the evaluated labels are the first player and the round winner. Since two player labels are evaluated in each of the 20 rounds, the metric is defined as:

$$A_{\text{player}} = \frac{N_{\text{correct players}}}{40}$$

- **Briscola recognition accuracy:** measures whether the briscola card is correctly recognized. The prediction is considered correct only if the predicted card matches the ground-truth briscola card. Since the briscola card is the same throughout, the metric is defined as:

$$A_{\text{briscola}} = \frac{N_{\text{correct briscola}}}{1}$$

- **Game result accuracy:** measures the correctness of the final game result. The evaluated fields are the final winner, the score of player North, and the score of player South. Since three result fields are evaluated, the metric is defined as:

$$A_{\text{result}} = \frac{N_{\text{correct result fields}}}{3}$$

Each metric is reported separately as the number of correct predictions over the total number of evaluated predictions for that category.

7 Delivery

The final delivery should contain the source code, build instructions, the generated per-round CSV outputs, output metrics for each game, and a short report describing the approach.